armfield

F1-16-MKII Impact of a Jet

Water is discharged vertically through a nozzle to strike a target carried on a stem, which extends through the cover. The dead weight of the moving parts are counterbalanced by a compression spring.

PRODUCT CODE: **F1-16-MKII**

Categories: Agricultural, Educational, F1-16-MKII Impact Of A Jet

Tags: Armfield , Compression Spring , Counterbalanced , Dead Weight , F1-16-MKII , Hemispherical Target , Impact Of A Jet , Momentum Theory , Nozzle , Vertical Force , Weights

DESCRIPTION

Water is discharged vertically through a nozzle to strike a target carried on a stem, which extends through the cover. The dead weight of the moving parts are counterbalanced by a compression spring. The vertical force exerted on the target plate is measured by adding the weights supplied to the weight pan.

The apparatus consists of a cylindrical clear acrylic fabrication with provision for levelling. Water is fed through a nozzle and discharged vertically to strike a target carried on a stem, which extends through the cover. A weight carrier is mounted on the upper end of the stem.

The dead weight of the moving parts is counterbalanced by a compression spring. The vertical force exerted on the target plate is measured by adding the weights supplied to the weight pan until the mark on the weight pan corresponds with the level gauge.

A total of eight targets are provided.

TECHNICAL SPECIFICATIONS

- Nozzle diameter: 8mm
- Distance between nozzle & target plate: 40mm
- Diameter of target plate: 36mm
- Target plates:
  - – 180° hemispherical target
  - – 120° target (cone)
  - – Flat target
  - – 30° target
  - – 60° target

- – Cup 135°
- – Oblique 30/150°
- – Oblique 45/135°

Requires Hydraulics Bench Service unit F1-10

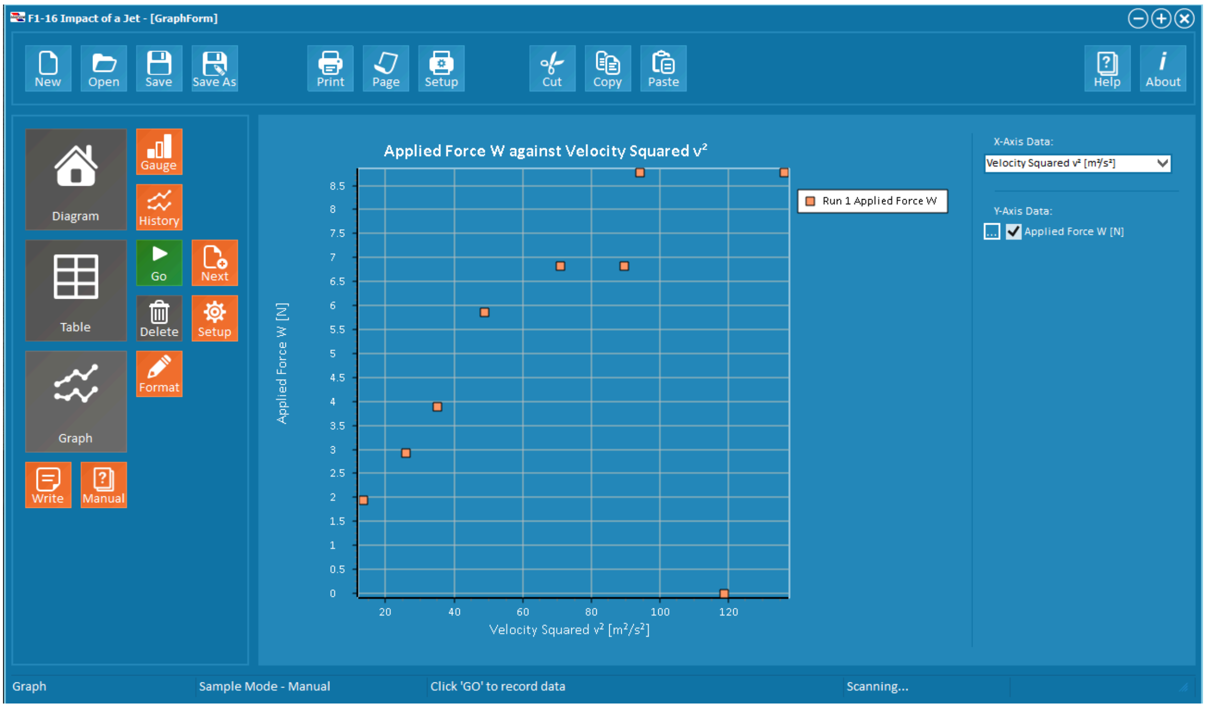
FEATURES & BENEFITS

Demonstration Capabilities

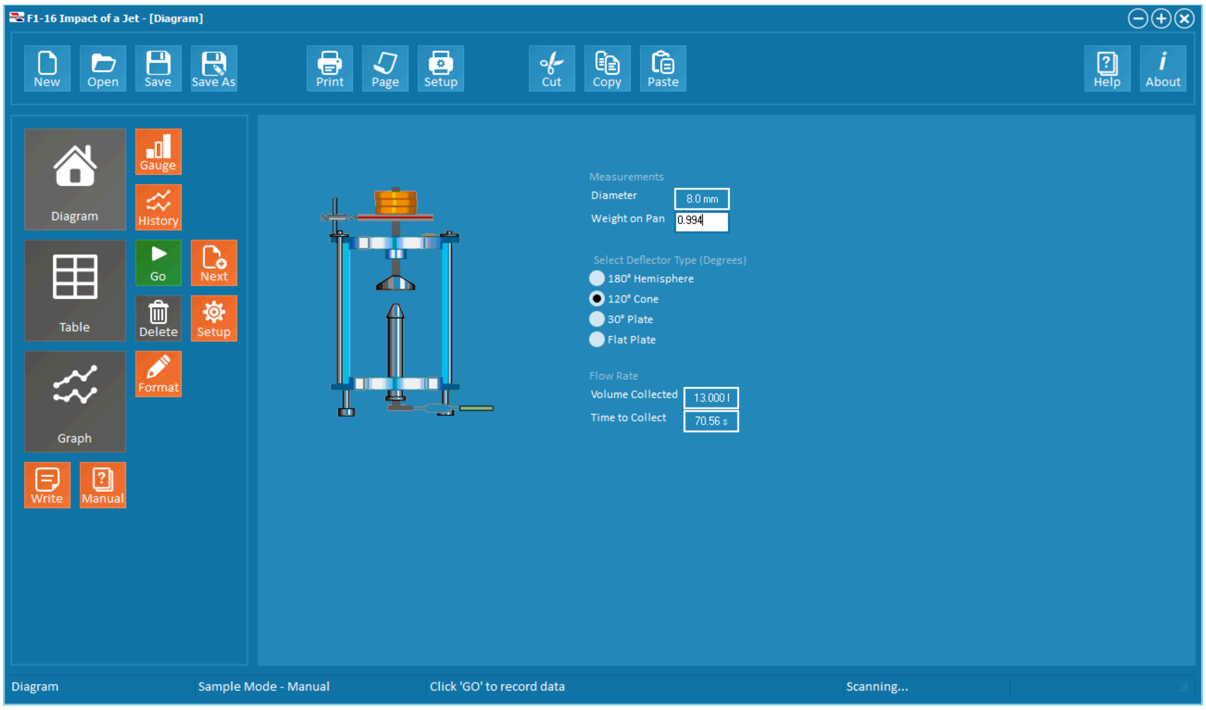
- Principle of linear momentum
- To investigate the reaction forces produced by the change in momentum of a fluid flow
- Measurement of the forces produced by a jet impinging on solid surfaces which produce different degrees of flow deflection

Practical Training Exercises

- Principle of linear momentum
- To investigate the reaction forces produced by the change in momentum of a fluid flow
- Measurement of the forces produced by a jet impinging on solid surfaces which produce different degrees of flow deflection



F1-16-MKII Impact of a Jet Graph



F1-16-MKII Impact of a Jet Diagram

Dimensions

Length: 0.325m

Width: 0.20m

Height: 0.50m

Order Codes

F1-16-MKII

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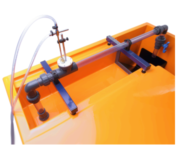

F1-33 PITOT TUBE DEMONSTRATOR

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